

25PY101-S2: Engineering Physics

Dr. Sreekar Guddeti

April 19, 2026

References

pre-M2T1 slot1

1 Topics

Topics	Avadhanulu	Neamen
Quantum Free Electron Theory (QFET)		
	35.15 QFET	
	35.16 density of states	3.4 Density of states
	35.17 carrier concentration	
	35.19 Failure of QFET	4.4.2 Complete ionization and freeze out
Fermi-Dirac distribution		
	31.4.2 Fermi-Dirac statistics	3.5 Statistical mechanics
	36.9 Energy band structure of conductor	
	31.4.2 (a) Electron concentration	
	31.4.2 (b) Heat capacity of metals	
Quantum Band Theory (QBT)		
	36.2 Band theory of solids (qualitative)	3.1 Allowed and forbidden bands
	36.4 Energy band structure	
	36.6 Energy band diagram	
	36.7 Classification of solids	
	36.11 Energy band structure of semiconductor	
	36.12 Electron and hole effective mass	
QBT in intrinsic s.c.		
	37.8 Carrier concentration	4.1 Charge carriers in s.c.
	37.9 Intrinsic carrier concentration	4.1.1 Equilibrium distribution of electrons and holes
		4.1.2 n_0 and p_0 equations
	37.11 Fermi level in i-s.c.	4.1.3 Intrinsic carrier concentration
	37.12 Intrinsic conductivity with temperature	4.1.4 Intrinsic Fermi level position
	37.13 Determination of band gap	
	37.14 Limitations of i-s.c.	

Table 1: Topics and corresponding sections in the course textbooks.

2 Test your understanding

QFET

1. What are the postulates of QFET.
2. Define Pauli's exclusion principle.
3. Define Fermi energy level.
4. Define density of states function.

Fermi-Dirac distribution

1. Explain the variation of Fermi-Dirac distribution function at $T = 0\text{ K}$ and $T \neq 0\text{ K}$.
2. Show that the probability of finding an electron of energy ΔE above the Fermi level is same as the probability of not finding an electron at energy ΔE below the Fermi level.
3. Show that the occupancy probabilities of two states whose energies are equally spaced above and below the Fermi energy add up to one.

Success and failures of QFET

1. If the Fermi level of metal is E_f , what is the fraction of thermalized electrons at temperature T .
2. A thermalized electron according to CFET increases the energy of metal by $\mathcal{O}(k_B T)$. However, according to QFET, the thermalized electron increases the energy of metal by $\mathcal{O}(f k_B T)$, where f is the fraction of thermalized electrons. Explain.
3. Prove that the dispersion relation $E - k$ in metals is given by

$$E(k) = \frac{\hbar^2 k^2}{2m}.$$

4. List the merits and demerits of QFET.

QBT

1. What are the postulates of QBT?
2. Define the periodic potential approximation.
3. In the parabolic approximation, since the conduction band is approximated by an upward parabola, the conduction band charge carrier is electron-like. Explain.
4. Hole has a negative effective mass. Explain.

QBT in intrinsic semiconductor

1. Discuss the variation of Fermi level position with temperature.
2. Discuss the variation of conductivity with temperature.

3 Technical terms

QFET

- Pauli's exclusion principle
- Fermi level
- Density of states
- Fermi velocity
- Quantum state
- energy level
- degeneracy
- multiplicity
- $E - k$ dispersion relation

F-D distribution

- occupied level
- unoccupied level
- probability of occupancy
- probability of vacancy

Success and failures of QFET

- specific heat capacity C
- ideal gas constant R
- thermalized electron
- $E - k$ dispersion relation

QBT

- periodic potential approximation
- forbidden energy level
- band structure
- conduction band
- valence band
- energy band gap
- energy band diagram
- conduction band minimum E_c
- valence band maximum E_v
- parabolic approximation
- effective mass of electron and hole

QBT in intrinsic semiconductor

- band structure
- zone folding
- hole
- conduction band effective density of states N_c
- conduction band effective density of states N_v
- intrinsic carrier concentration n_i
- midgap energy level E_{midgap}

End of document